1. Competency #9: Translate a problem into a standard mathematical representation or algorithm

> *Translate a problem into a standard mathematical representation or algorithm*

Highly relevant (⭐⭐⭐⭐⭐)
Students must convert the verbal recipe into a **mathematical model**, such as:

- A proportion: oil constitutes $\frac{60}{100} = 0.6$ (or 60%) of the total;
- Or a scaling factor: since 150 mL is $\frac{150}{100} = 1.5$ times the original, oil needed = $60 \times 1.5 = 90$ mL.

This process—transforming a real-world scenario (making dressing) into a **standard mathematical operation (proportion, multiplication)**—directly aligns with Competency #9.

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2. Competency #1: Select a mathematical description or representation that captures the problem

> *Select a mathematical description or a representation that describes a problem*

Highly relevant (⭐⭐⭐⭐☆)
The solver must choose an appropriate modeling approach:

- Unit rate: 0.6 mL oil per 1 mL of sauce → 150×0.6 ;
- Scaling factor: multiply all ingredients by 1.5;
- Fractional part: oil is $\frac{3}{5}$ of the mixture.

While different representations lead to the same answer, the act of **selecting a suitable mathematical model** reflects deep understanding of the problem structure—exactly what Competency #1 emphasizes.

3. Competency #6: Identify and describe the mathematical elements of a real-world situation, including key variables

> *Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables*

Highly relevant (⭐⭐⭐⭐☆)
Key variables include:

- Total volume (100 mL → 150 mL);
- Individual ingredient volumes (oil, vinegar, soy sauce);
- The invariant ratio among components (implicit constraint: flavor consistency requires fixed proportions).

Recognizing that "keeping the taste the same" means **maintaining constant ratios** is the mathematical essence of the problem. Missing this would lead to incorrect solutions.

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4. Competencies #2, #10, #11: Identify key variables / Use mathematical tools (symbols, variables) to describe relationships

> *Identify the key variables... Use mathematical tools (variables, symbols, diagrams) to describe relationships*

Moderately relevant (⭐⭐⭐☆☆)
Although often solved mentally due to simple numbers, a full modeling approach might introduce variables:

- Let x = required oil volume → $\frac{x}{150} = \frac{60}{100}$;
- Or define scaling factor $k = \frac{150}{100}$, so $x = 60k$.

Even if not required to write equations, **identifying variables (oil amount, total volume) and their relationship** is foundational to modeling—consistent with Competencies #2, #10, and #11.

5. Competency #8: Recognize connections between the problem and known mathematical concepts or procedures

> *Recognise aspects of a problem that correspond with known problems or mathematical concepts*

Moderately relevant (⭐⭐⭐☆☆)
Students should recognize this as a **proportional reasoning** problem—a common type in elementary/middle school (e.g., recipe scaling, map scales). This pattern recognition allows them to quickly apply familiar solution strategies (e.g., cross-multiplication), aligning with Competency #8.

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6. Competency #3: Select a representation appropriate to the problem context

> *Select a representation appropriate to the problem context*

Low relevance (⭐⭐☆☆☆)
The context is straightforward, and a simple arithmetic proportion suffices. While one could use tables or bar models for visualization, they are unnecessary. Thus, the choice of representation plays a minor role.

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7. Competencies #4, #5, #7, #12: Decode information / Recognize patterns / Simplify / Identify assumptions

> *Read/decode tasks... Recognise patterns... Simplify... Identify constraints/assumptions*

1. 第9条：将问题转化为标准数学表示或算法

> *Translate a problem into a standard mathematical representation or algorithm*

高度相关（⭐⭐⭐⭐⭐）
学生需将文字描述的配方转化为**比例模型**或**分数/倍数关系**。例如：

- 原配方中油占总量的 $\frac{60}{100} = 0.6$ （即60%）；
- 或理解为“每100 mL酱料需60 mL油”，因此150 mL需 $60 \times \frac{150}{100} = 90$ mL。

这正是将现实问题（做酱料）**翻译为标准数学操作（比例、乘法）**的过程，完全契合第9条。

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2. 第1条：选择能描述问题的数学表述或模型

> *Select a mathematical description or a representation that describes a problem*

高度相关（⭐⭐⭐⭐☆）
解题者需主动选择合适的数学模型：是用**比例**、**单位率（unit rate）**、**分数**，还是**线性缩放因子**？例如：

- 单位率：每1 mL酱料需0.6 mL油 → 150 × 0.6；
- 缩放因子：150 ÷ 100 = 1.5倍 → 60 × 1.5。

选择哪种表述虽不影响结果，但体现了对问题结构的理解和建模起点的选择，符合第1条。

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3. 第6条：识别并描述现实问题中的数学要素，包括关键变量

> *Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables*

高度相关（⭐⭐⭐⭐☆）
关键变量包括：

- 总体积（100 mL → 150 mL）；
- 各成分体积（油、醋、酱油）；
- 成分间的固定比例关系（隐含约束：比例不变）。

学生必须意识到“保持口味一致”意味着**比例恒定**，这是问题的数学本质。忽略这一点（如误以为只增加油）就会出错。

4. 第2条 & 第10/11条：识别关键变量 / 使用数学工具（符号、变量等）描述结构

> *Identify the key variables... Use mathematical tools (variables, symbols, diagrams) to describe relationships*

中等相关（⭐⭐⭐☆☆）
虽然本题数值简单，常被心算解决，但完整的建模过程可引入变量：

- 设所需油量为 x，则 $\frac{x}{150} = \frac{60}{100}$；
- 或定义缩放因子 $k = \frac{150}{100}$，则 $x = 60k$。

尽管题目不要求写出公式，但**识别变量（油量、总量）及其关系**是建模基础，符合第2、10、11条。

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5. 第8条：识别问题中与已知数学概念或程序的对应部分

> *Recognise aspects of a problem that correspond with known problems or mathematical concepts*

中等相关（⭐⭐⭐☆☆）
学生需识别这是**比例问题**（proportional reasoning），属于小学/初中常见题型（如食谱缩放、地图比例尺）。这种“模式识别”能力帮助快速调用已有解题程序（如交叉相乘），符合第8条。

6. 第3条：选择适合问题情境的表示方式

> *Select a representation appropriate to the problem context*

低相关（⭐⭐☆☆☆）
本题情境简单，通常直接用**算术比例**即可，无需复杂表示（如图表、方程组）。虽然可画表格或条形图辅助理解，但非必需，故相关性较低。

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7. 第4、5、7、12条：解读信息 / 识别结构 / 简化问题 / 识别假设

> *Read/decode tasks... Recognise patterns... Simplify... Identify constraints/assumptions*